

Fully integrated hydrological modelling
MIKE SHE

Dr. Subramani Ravi



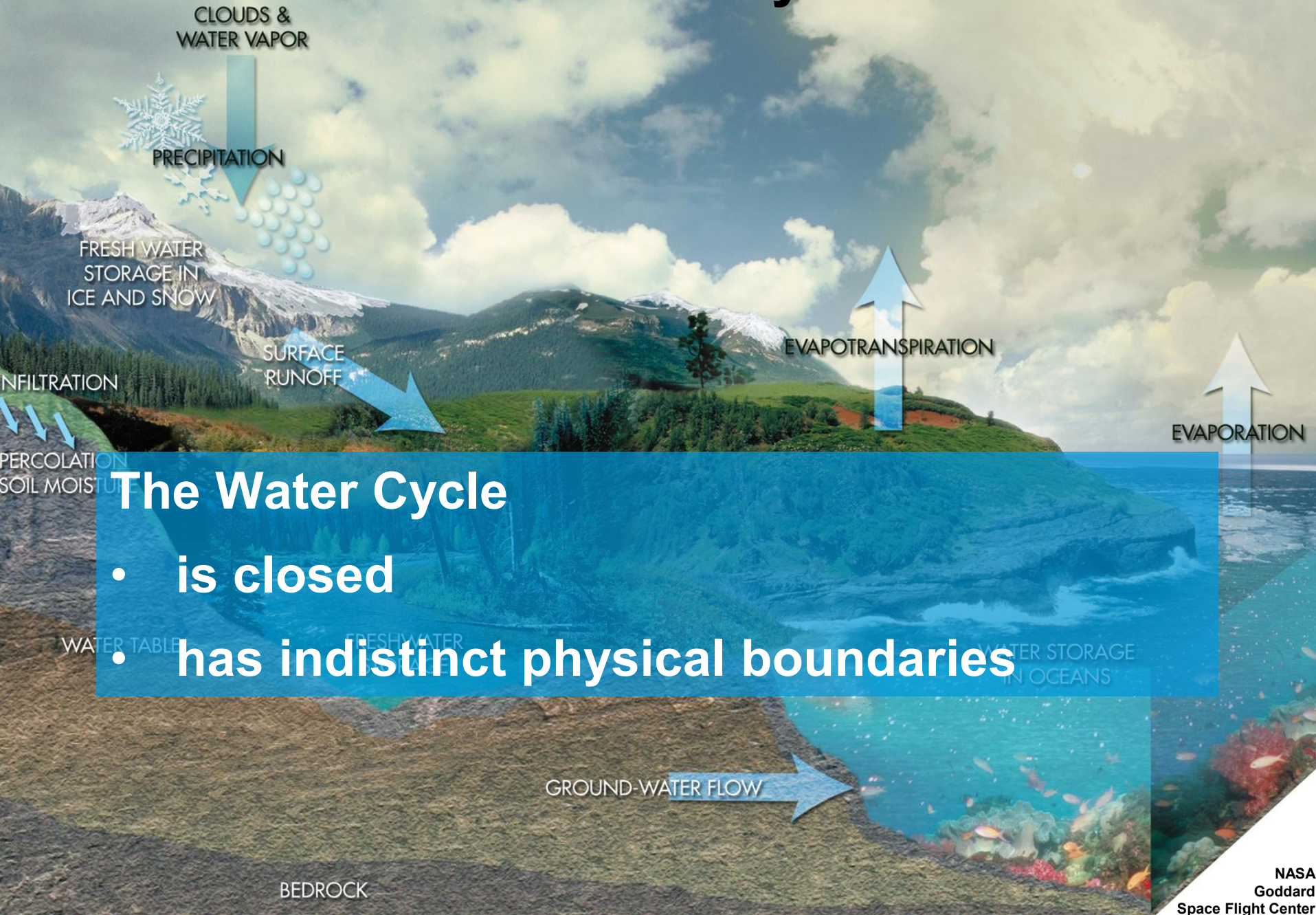


Analyse groundwater, surface water, recharge
and evapotranspiration processes with
MIKE SHE, the most advanced and flexible
framework for integrated hydrological modelling

01.

Introduction to integrated hydrology

The Water Cycle

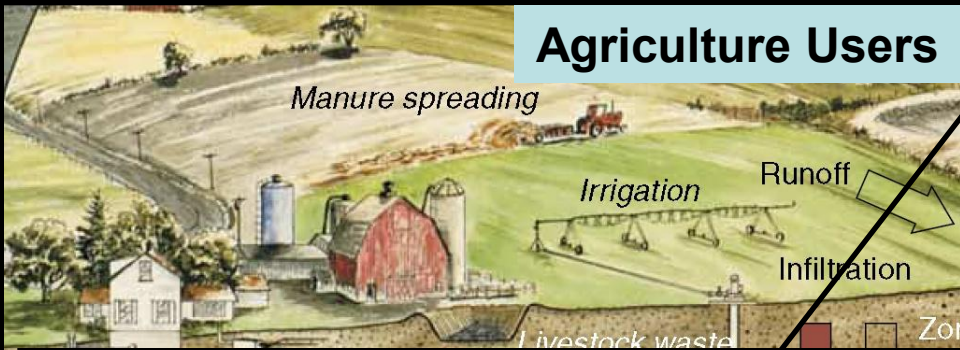


The Water Cycle

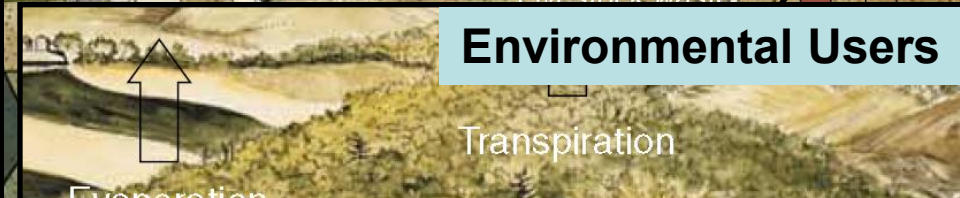
- is closed
- has indistinct physical boundaries

Water Users

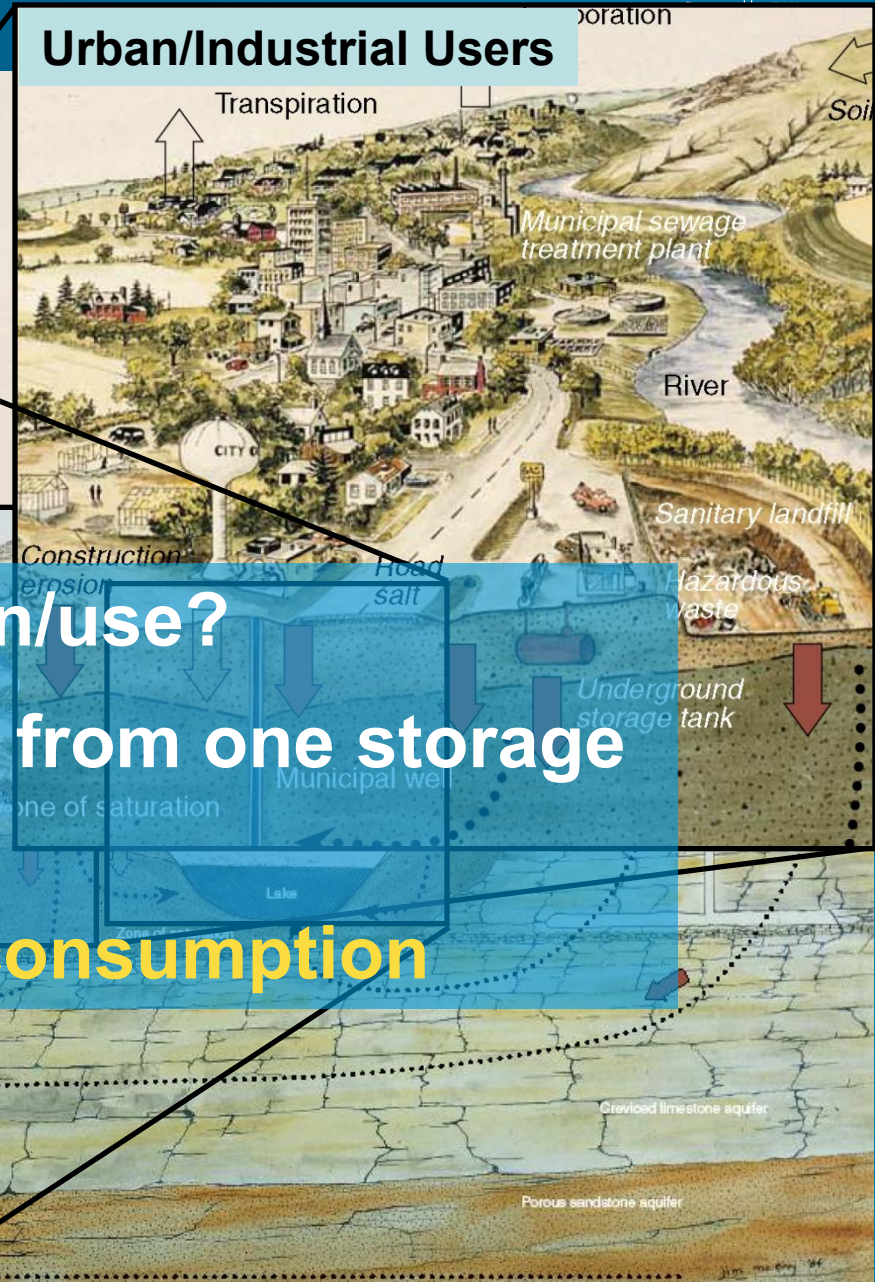
Agriculture Users



Environmental Users



Urban/Industrial Users



What is water consumption/use?

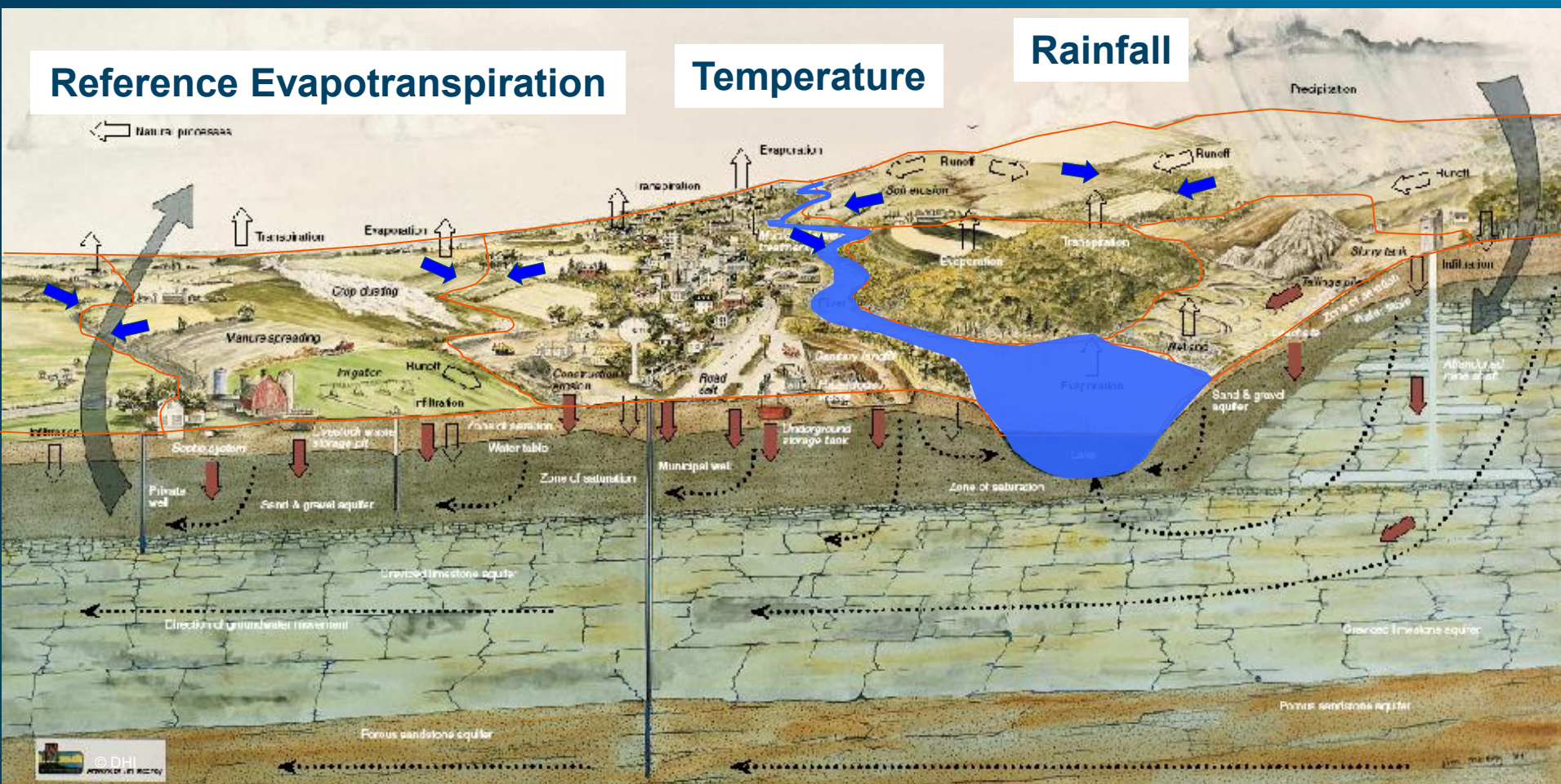
- The movement of water from one storage to another

ET is the only true consumption

Environmental Modelling

Water Modelling Hydrologic Cycle

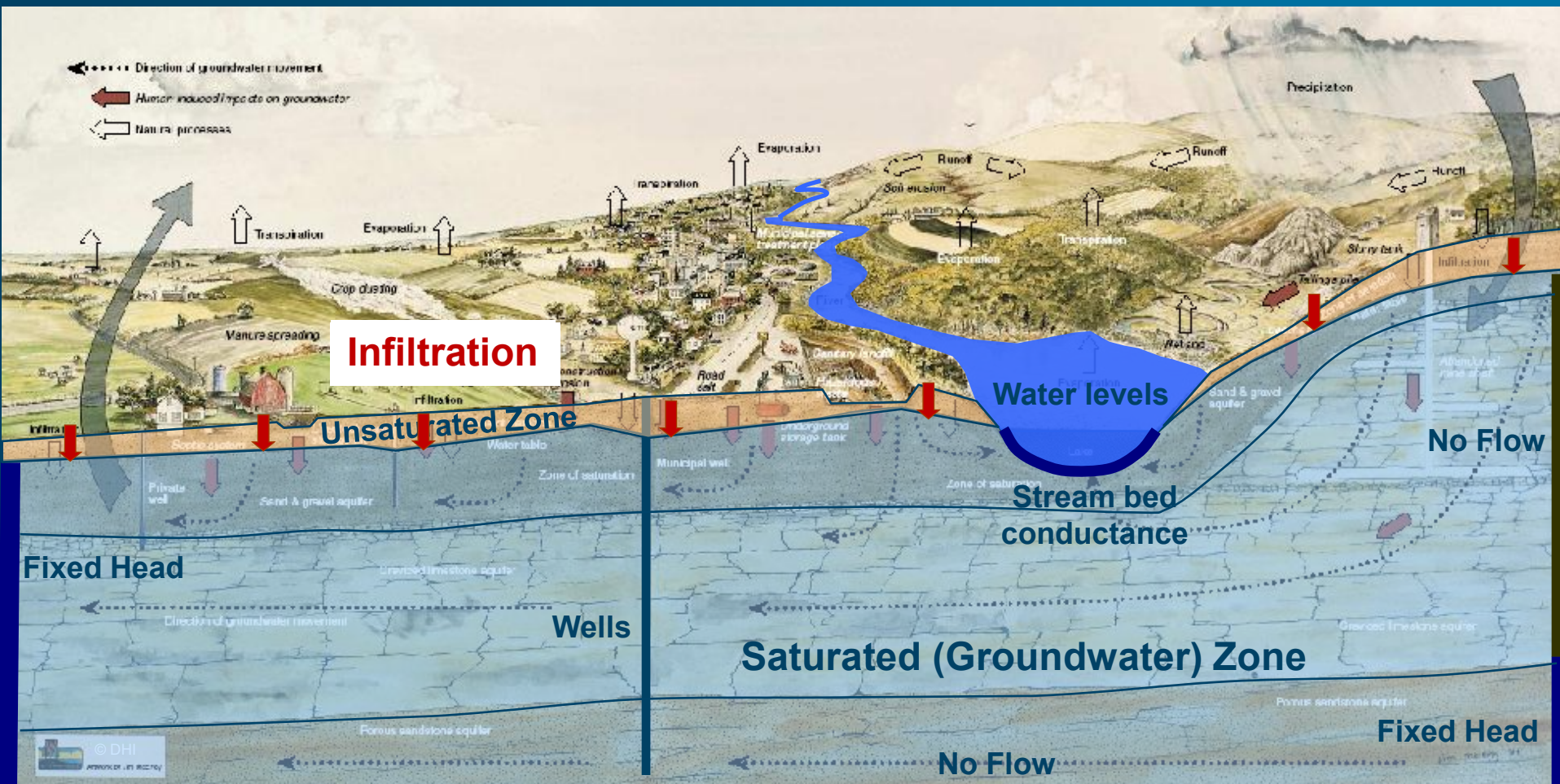
❓ Rivers and Lakes - Well established processes and models



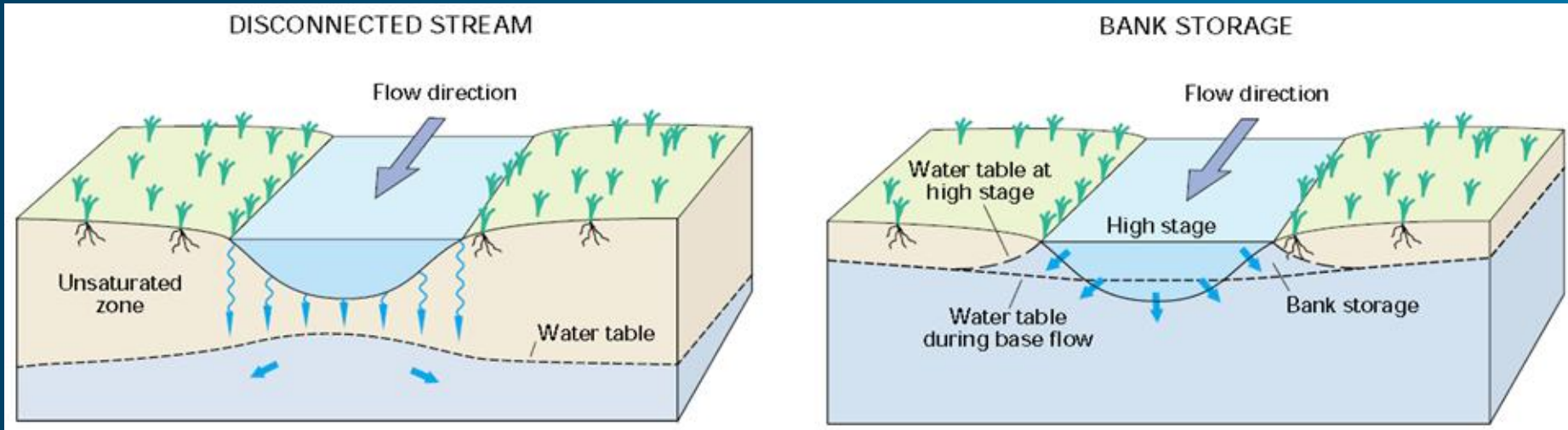
Environmental Modelling

Water Modelling ? Hydrologic Cycle

? Groundwater - Well established processes and models



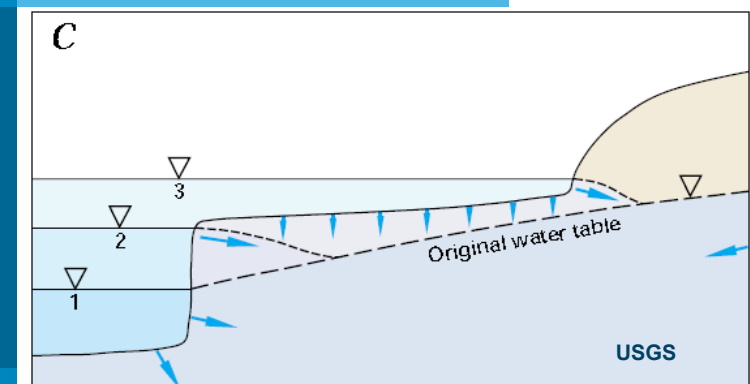
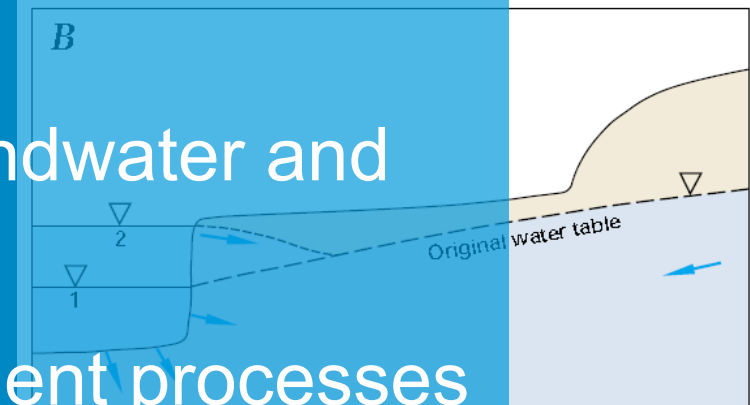
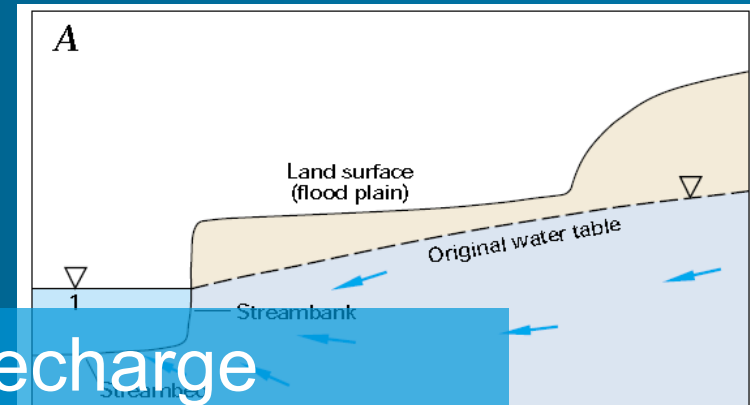
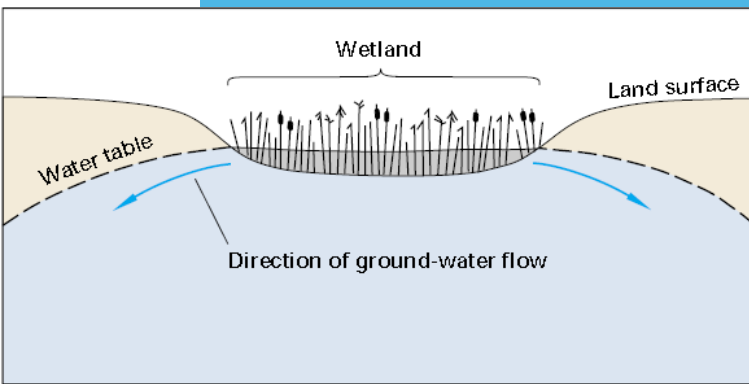
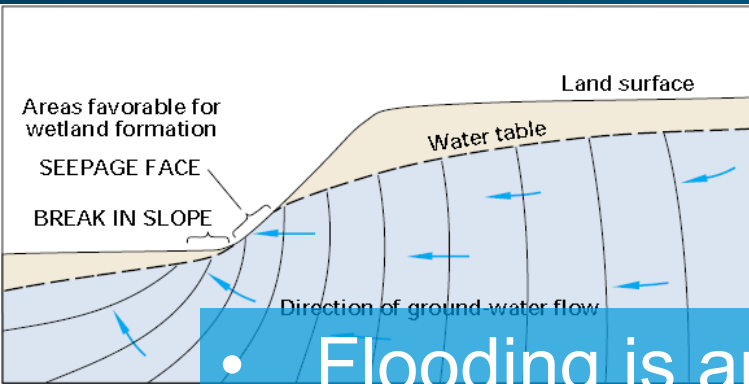
Groundwater – surface water interaction is complex



- Baseflow varies non-linearly with flow
- Stream levels are dynamic
- Stream levels depend on runoff
- Stream levels depend on drainage



Groundwater – surface water interaction is complex



- Flooding is an important recharge process
- Wetlands are where groundwater and surface water intersect
- Both are critical to catchment processes

Runoff

Infiltration + ET

~~Runoff~~ depends on land surface

- Land use, vegetation, soil type
- Elevation, topography, slope
- Drainage network (e.g. surface ponds)

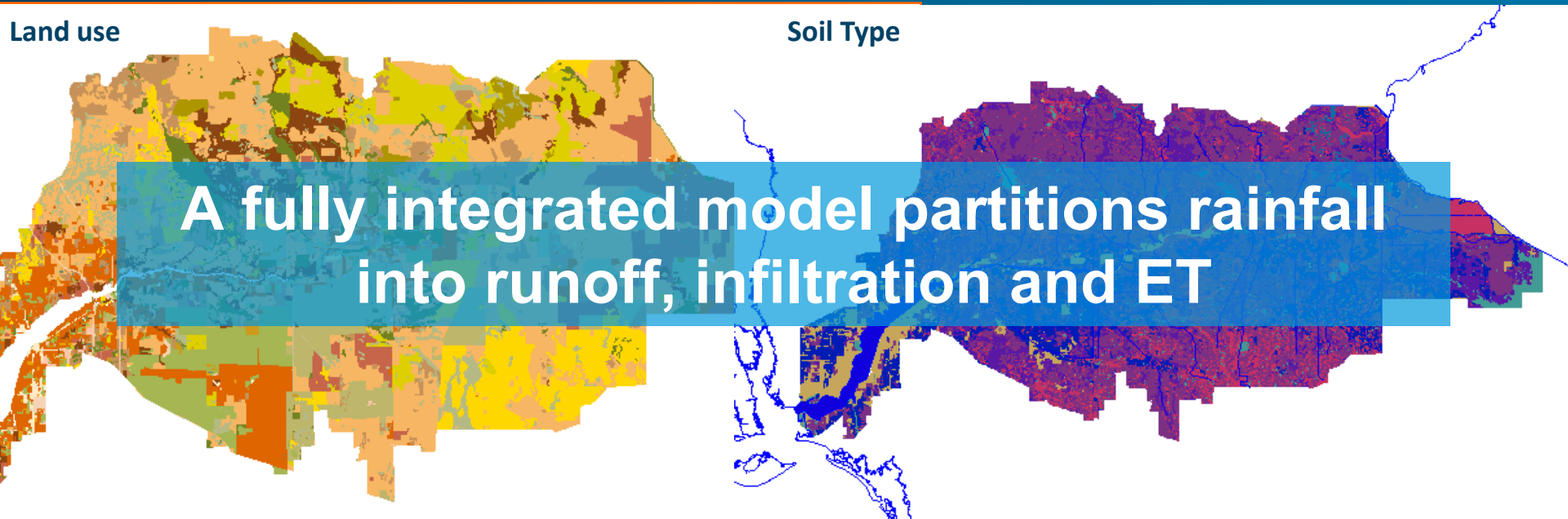
Infiltration + ET

~~Runoff~~ depends on meteorology

- Precipitation type - rain, snow
- Intensity, duration, spatial distribution, prior amount
- Air temperature, wind, relative humidity, etc.

Land use

Soil Type

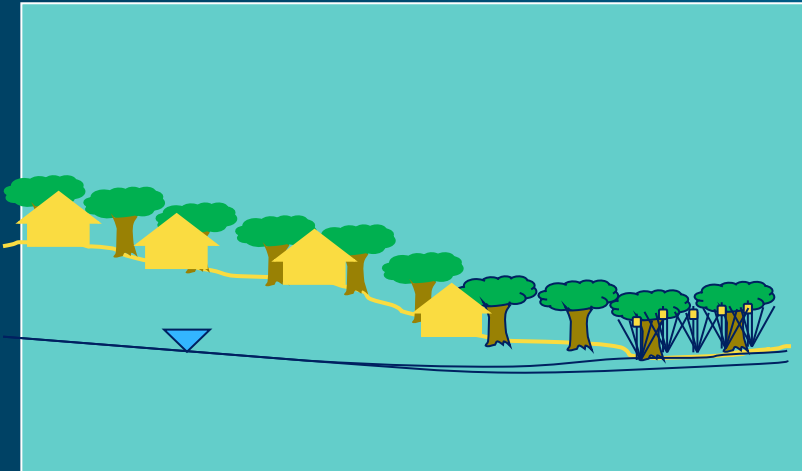


A fully integrated model partitions rainfall into runoff, infiltration and ET

Environmental Modelling

Why is dynamic feedback important?

Urbanization



Deforestation

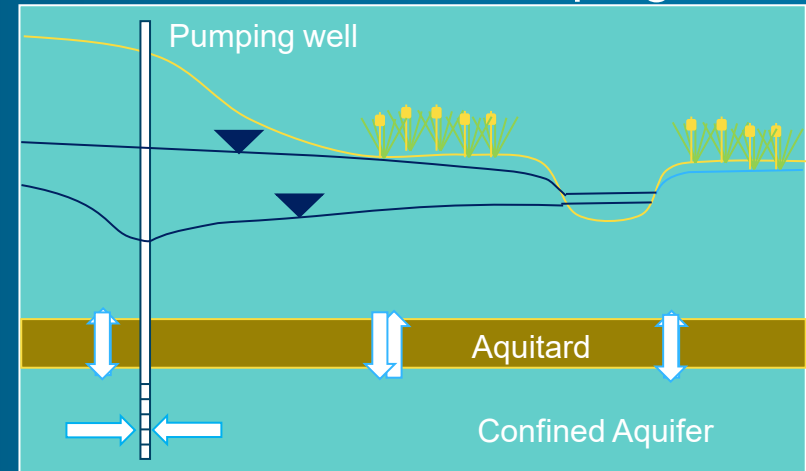
Urbanization

Increased runoff to low area

Wetland formation

Changes to vegetation and wildlife habitat

Groundwater Pumping



Install pumping well

Depressurize confined aquifer

Dewater wetland and reduce baseflows to river

Changes to vegetation and aquatic habitat

02.

MIKE SHE – an overview

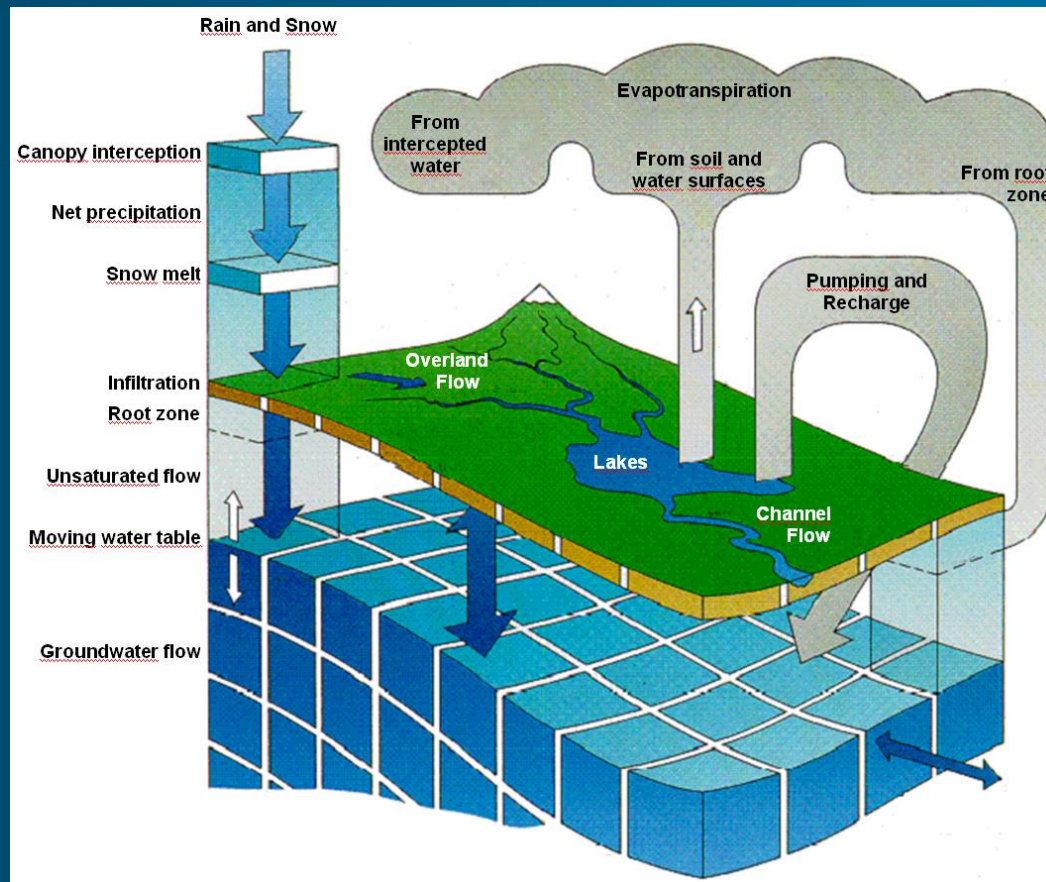
MIKE SHE – Integrated catchment modelling

Groundwater and Surface Water One Water — One Resource — One Model Integrated water quality

Precipitation
and snowmelt

Vegetation based
evapotranspiration
and infiltration

Unsaturated
groundwater
flow



Demand
driven
irrigation

Overland
surface flow
and flooding

Channel flow
in rivers and
lakes
(MIKE +)

Saturated groundwater flow

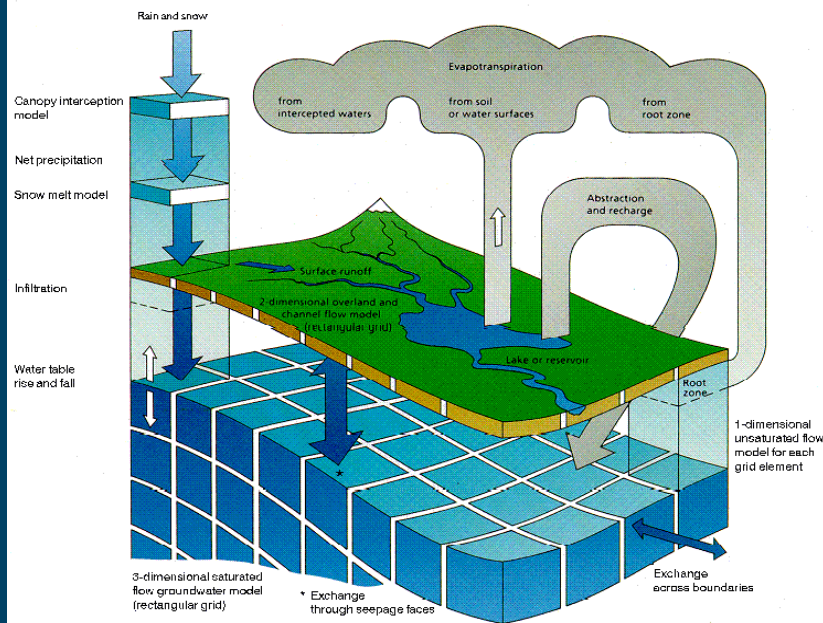
The SHE story

Systeme Hydrologique Europeen

- Consortium from mid 1970s
- 1st commercial version mid 1980s

MIKE SHE

an Integrated Hydrological Modelling System



MIKE SHE up to 1990

Physically based process descriptions
= conservation of mass and momentum
= physically meaningful model parameters

CHANNEL FLOW

⇒ 1D diffusive wave / Full HD

OVERLAND FLOW

⇒ 2-D diffusive wave / Full HD

EVAPOTRANSPIRATION

⇒ Kristensen & Jensen

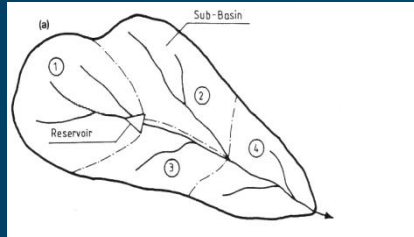
UNSATURATED ZONE FLOW

⇒ Full 1D Richards Equation

SATURATED ZONE FLOW

⇒ 3D Darcy flow

Framework for hydrologic modelling



**Parameters
specified by
catchment**

Lumped

**Conservation
of mass and
Momentum
e.g. Darcy's Law**

Pro: Does not require catchment data

Con: Relies heavily on calibration

(e.g. discharge)

Con: Not suited for scenario analysis

Conceptual

**Catchment
water budget**

Physics based

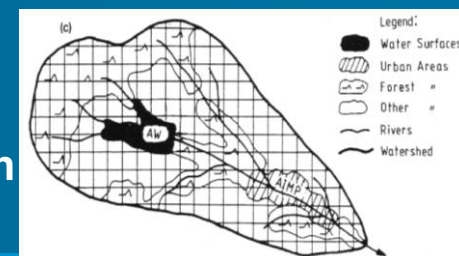
**Pro: Uses available catchment data, thus
less dependent on calibration**

Pro: Well suited for scenario analysis

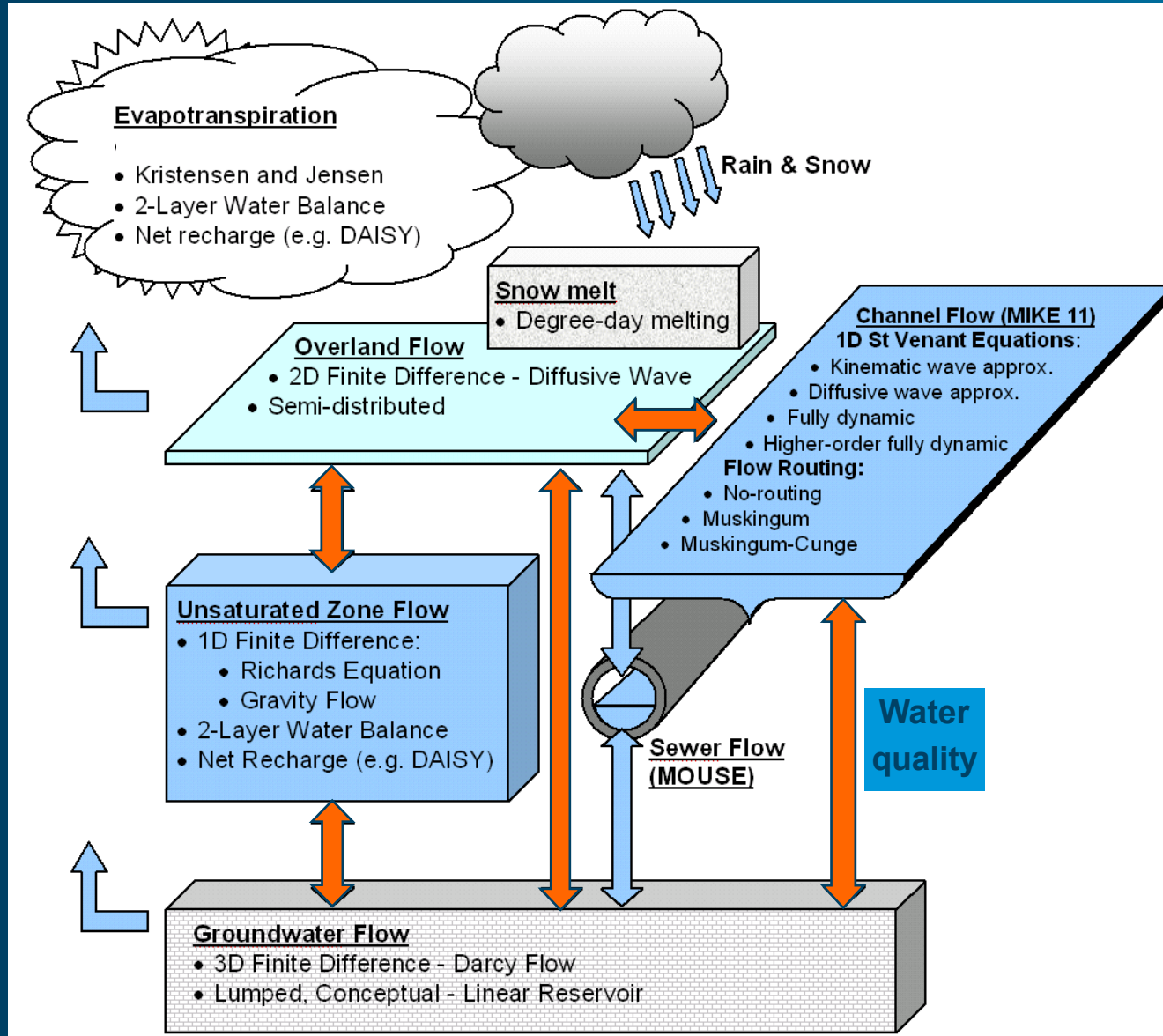
Con: Often not enough data

Distributed

**Parameters
specified on
grid**



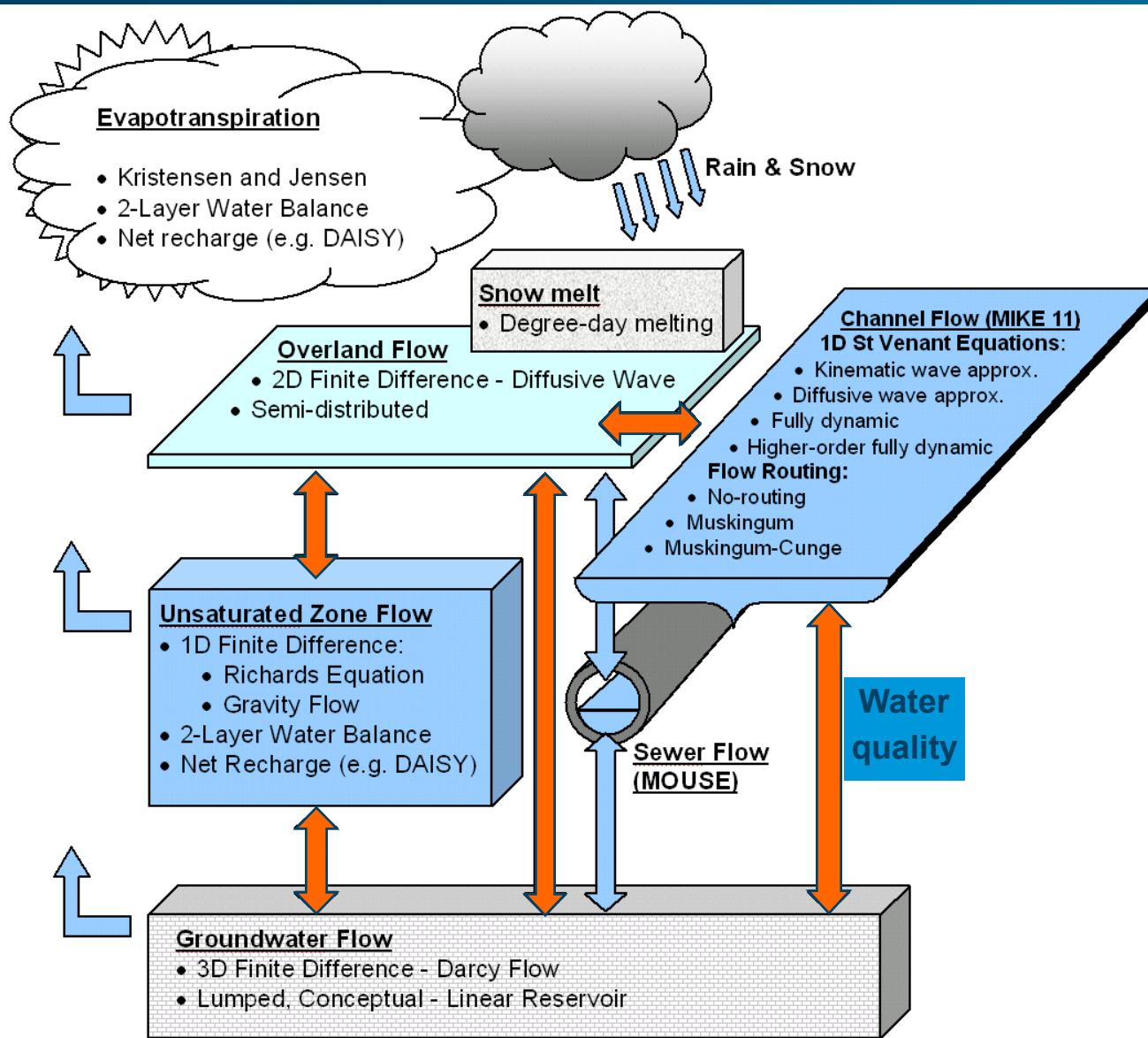
MIKE SHE – flexible process descriptions



Physics based methods often not required for every processes

Simpler methods faster and require less data

MIKE SHE – flexible process descriptions



Different

- Spatial scales
- Time scales

Processes mixed as required

Explicit coupling

Time scales independent and automatically controlled

MIKE SHE – Integrated catchment modelling

Dynamic, distributed hydrology model

Flexible spatial and time scales

Water flow and quality

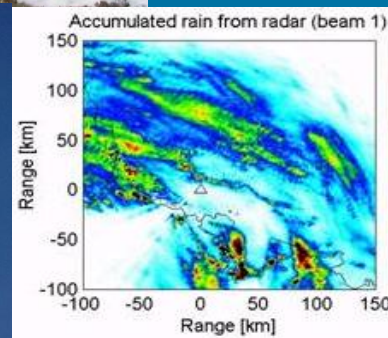
Applications:

- Wetland management
- Conjunctive water use
- Climate and land use change impacts
- Catchment nutrient balances
- Irrigation management
- Drought and flood planning
- Urban drainage and groundwater interaction
- Environmental river flows

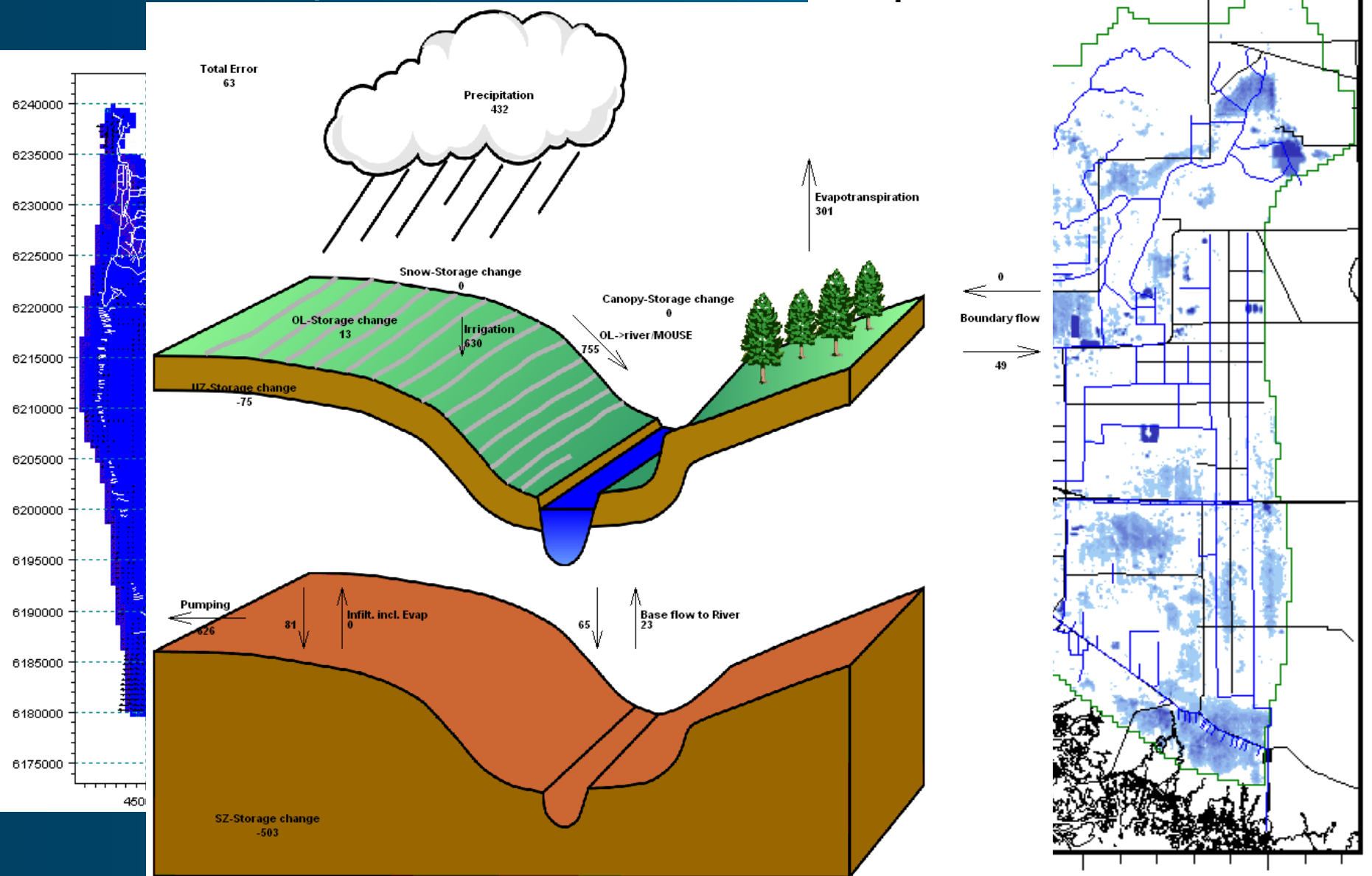
Basically:

How and where does ALL the water flow?

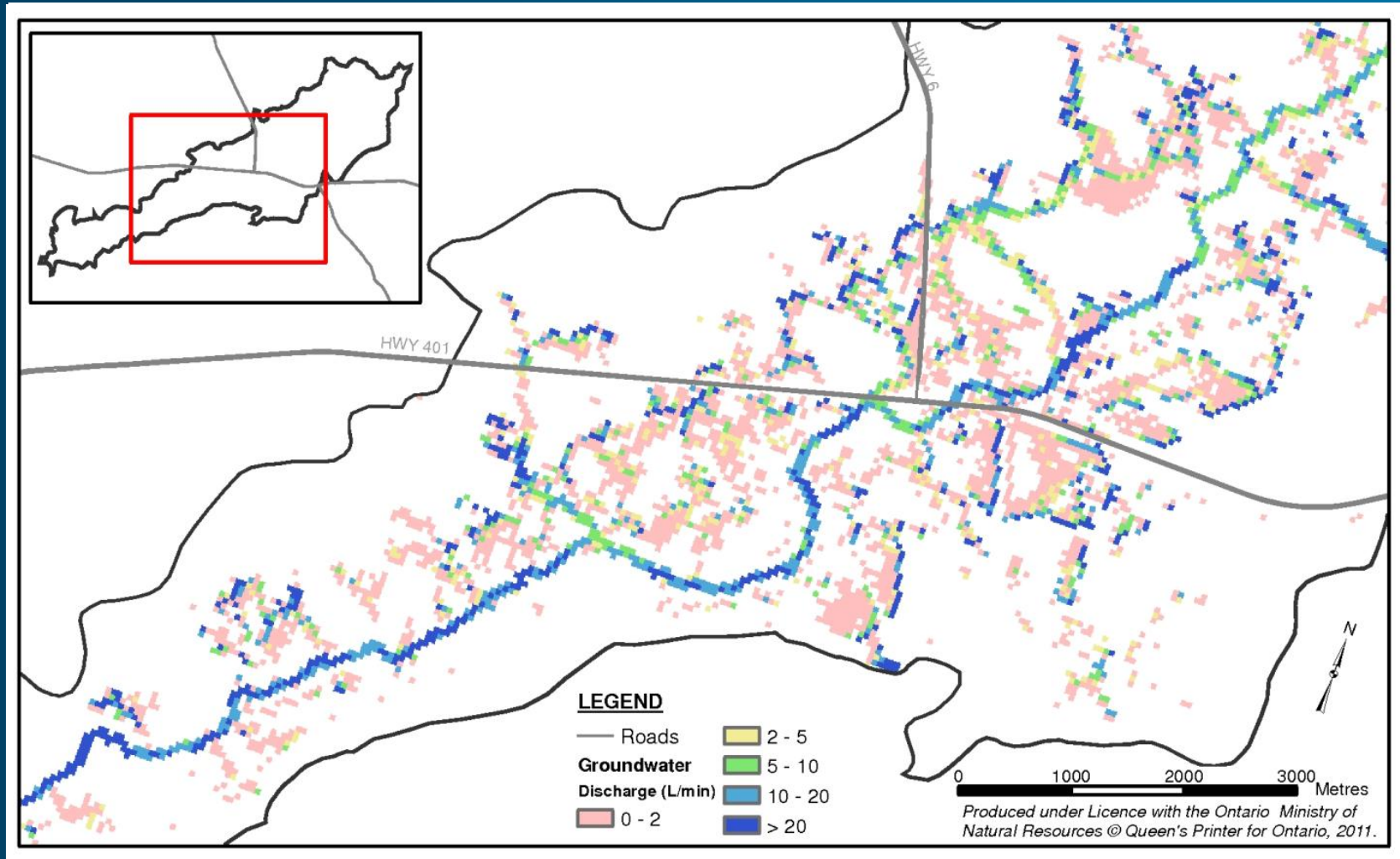
When will it get there and what will it be like?



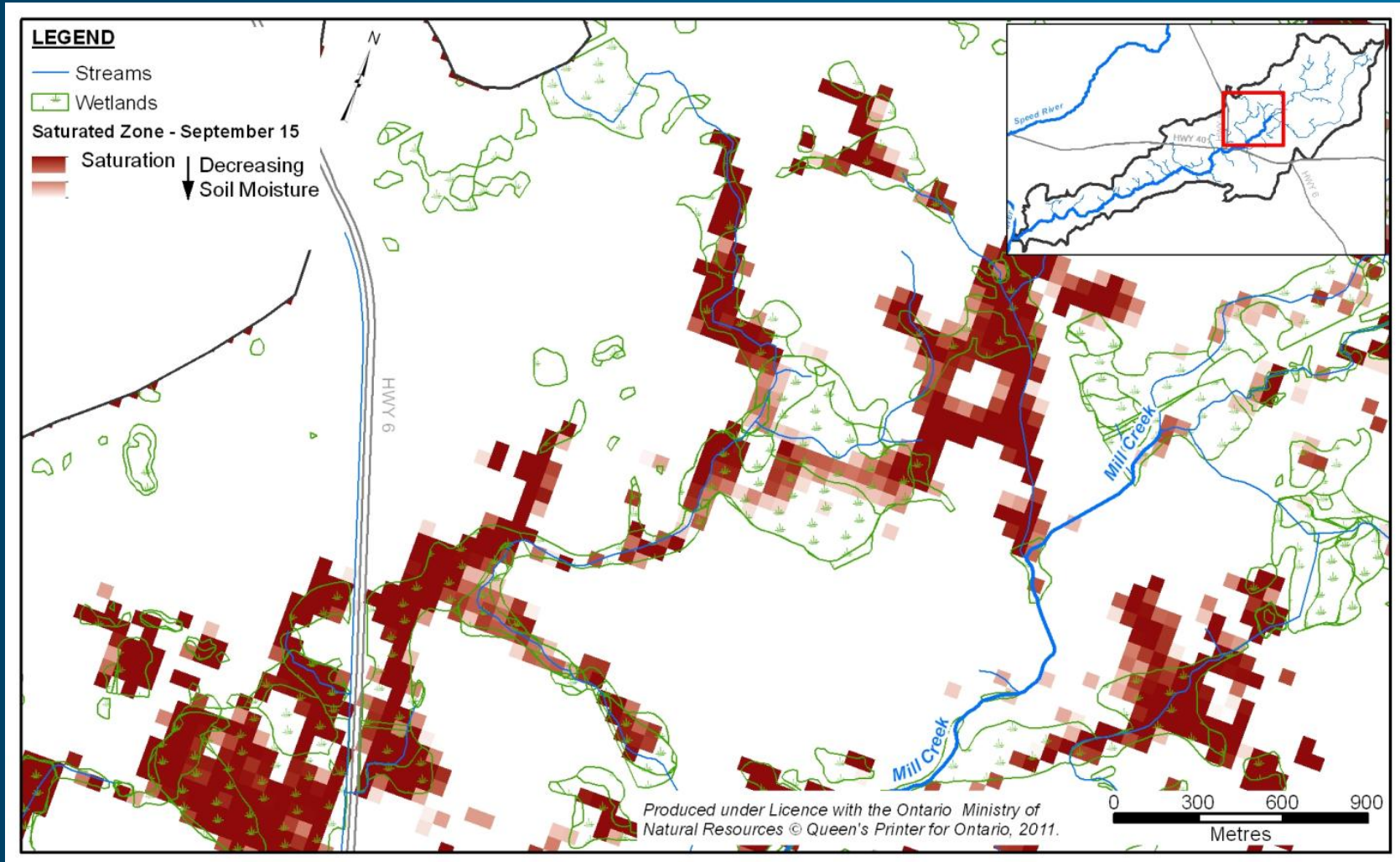
Potential Outputs



Groundwater discharge areas



Seasonal wetland deliniation



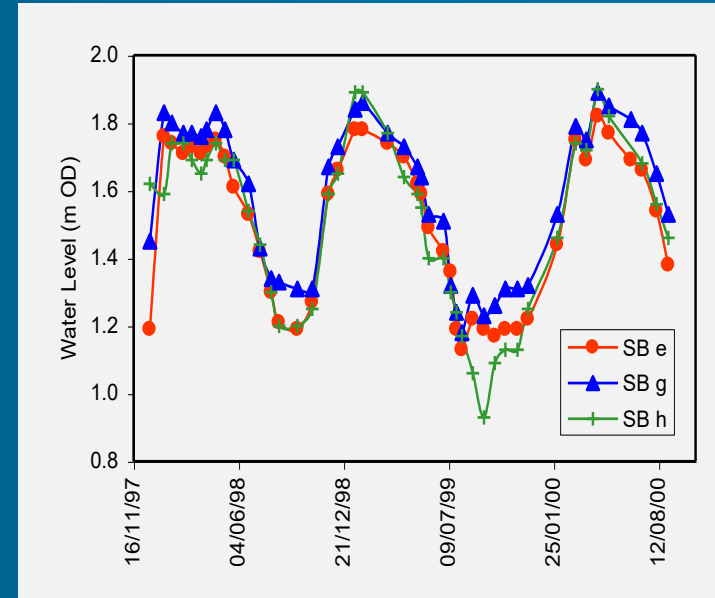
03.

Data Requirements for integrated modelling

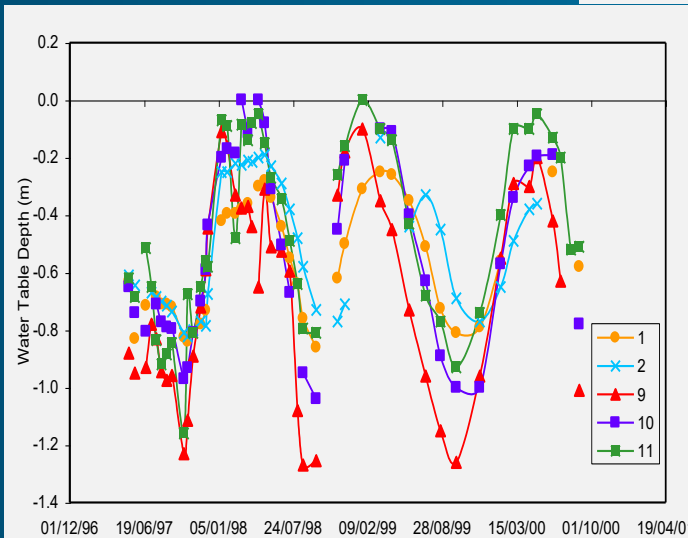
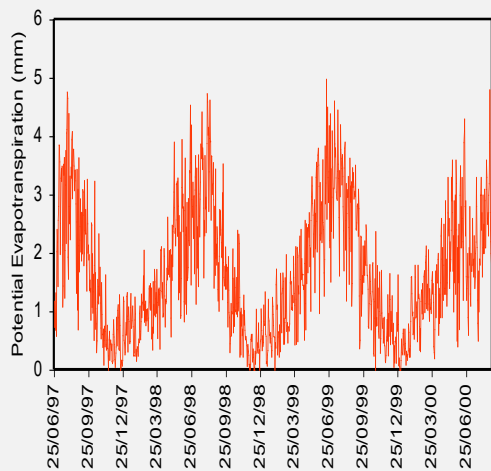
Temporal Data



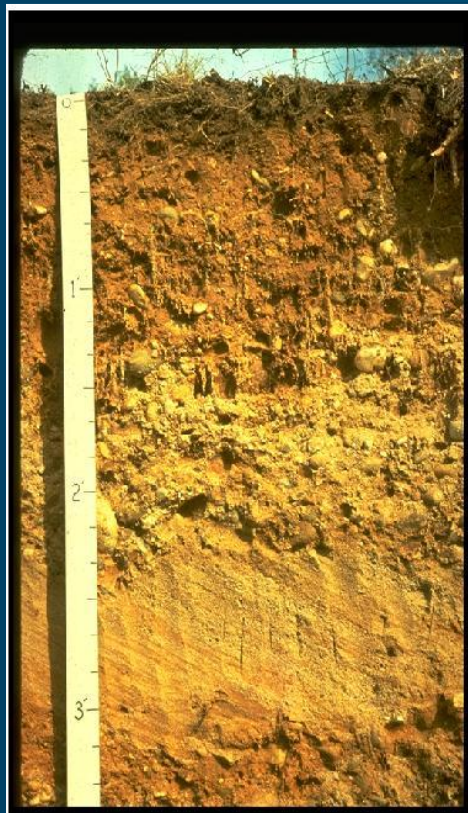
Forcing Functions



Calibration
Observations



Property Data



Ap, 10YR3/2,
Sandy loam.

Bw1, 10YR 5/8,
gravelly sandy
loam.

Bw2, 10YR 5/6,
gravelly loamy
sand.

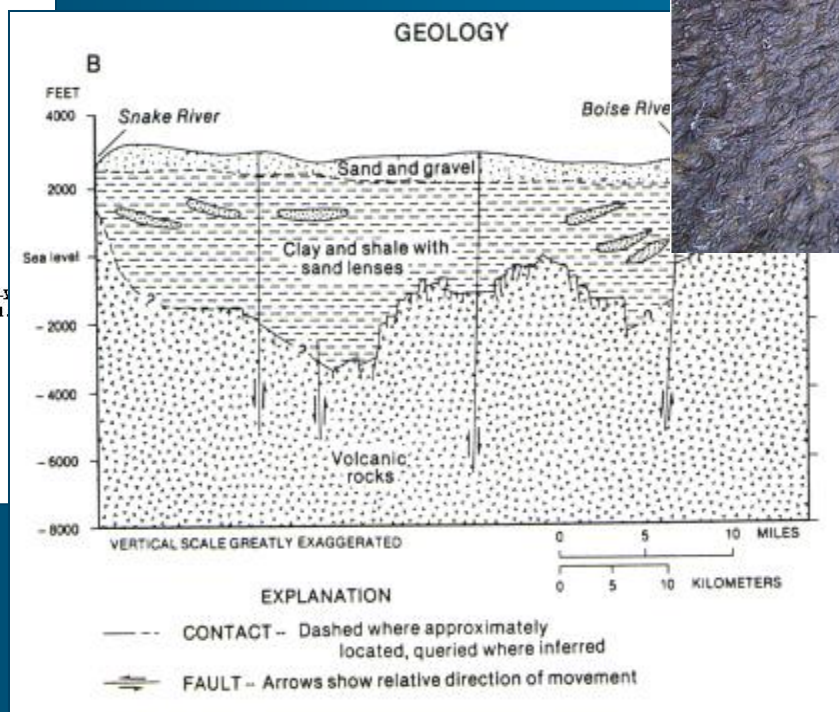
C1, very
gravelly
coarse sand.

C2, sand.

C3, gravelly
coarse sand.

Profile of a Hickley soil series. Hickley soils are excessively drained soils formed in glacial fluvial deposits of stratified sand and gravel.

Photo by P. Fletcher (USDA-NRCS)



Data requirements - groundwater

Physical properties

- Hydraulic conductivities
- Specific yield and storage
- Subsurface geometry

Hydraulic boundaries

- Flow and head boundaries
- Pumping rates
- Groundwater recharge

Data requirements - streams

Physical properties

River bed roughness

River geometry

Hydraulic boundaries

Surface runoff

River water levels/inflows

Data requirements

Groundwater + streams + unsaturated zone

Physical properties

Hydraulic conductivities

Specific yield and storage

Subsurface geometry

River bed roughness

River geometry

River bed conductance

Soil properties (e.g. θ_s , θ_{fc} , K_s)

Topography

Hydraulic boundaries

Flow and head boundaries

Pumping rates

Surface infiltration

Surface runoff

River water levels/inflows

Data requirements

Groundwater + streams + unsaturated zone + **land surface**

Physical properties

Hydraulic conductivities

Specific yield and storage

Subsurface geometry

River bed roughness

River geometry

River bed conductance

Soil properties (e.g. θ_s , θ_{fc} , K_s)

Topography

Runoff properties (e.g. roughness)

Vegetation properties (e.g. LAI, root depth)

Hydraulic boundaries

Flow and head boundaries

Pumping rates

Precipitation

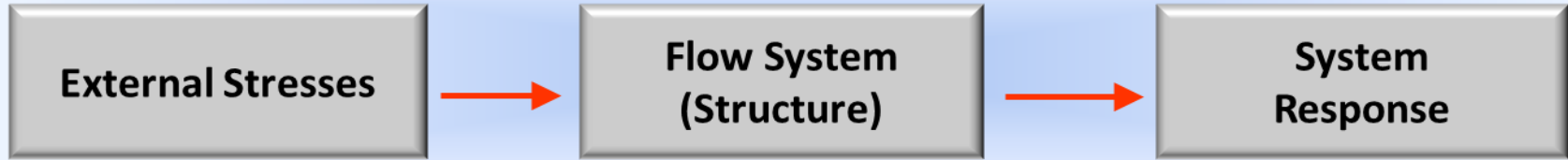
Evapotranspiration

River water levels/inflows

An integrated
hydrologic model
internalizes
“boundary conditions”

Data Needs

Flow Model



• Climate Data:

- Rainfall
- Reference ET
 - Wind
 - Air Temperature
 - Humidity
 - Solar Radiation
- Groundwater Pumping
 - Irrigation
 - Dewatering
- Artificial Recharge
- River Operations

Subsurface Flow System

- Geology
- Stratigraphy,
- Surface Soils
- Topography
- Hydraulic Properties,
- Subsurface Utilities

Surface Water Flow System

- Natural Drainage Profiles and Sections
- Drainage Diversions/Berms
- Hydraulic Structures - Dams, Weirs, Gates, others

Subsurface

- Groundwater Heads/Gradients
- Groundwater Flows
- Infiltration/Recharge
- Groundwater Discharge to Surface Water - Seepage
- Soil Moisture w/depth & time

Surface Water

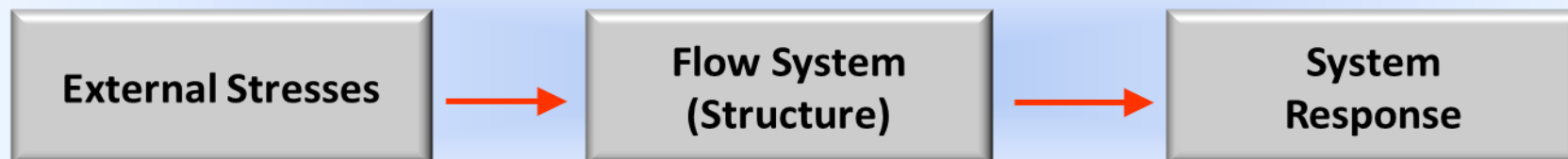
- Overland Flow/Stage
- Surface Flow/Stage
 - Drainages
 - Structures
- Baseflows (Gaining/Losing)

Rainfall is the primary source of water,

Evapotranspiration is the primary sink of water

- Pond/Reservoir levels
- Infiltration Rates
- Evapotranspiration

Flow Model



• Climate Data:

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 - Humidity
 - Solar Radiation

• Groundwater Pumping

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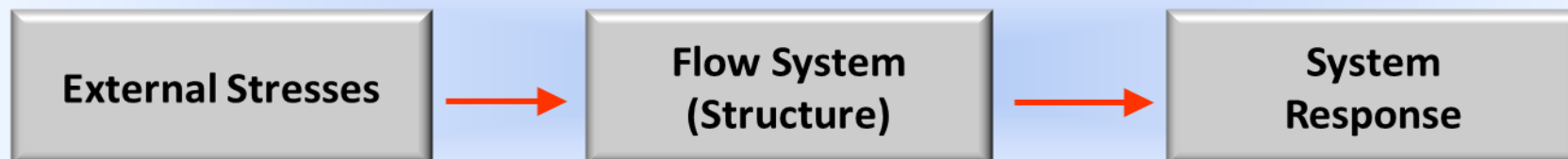
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Surface Water

- Overland Flow/Stage
- Surface Flow/Stage
 - Drainages
 - Structures
- Baseflows (Gaining/Losing)

Water pumping is the primary internal transfer of water between various storages

Flow Model



- *Climate Data:*

- Rainfall
- Reference ET
 - Wind
 - Air Temperature
 - Humidity
 - Solar Radiation

- *Groundwater Pumping*

- Irrigation
- Dewatering

- *Artificial Recharge*

- *River Operations*

Subsurface Flow System

- *Geology*
- *Stratigraphy,*
- *Surface Soils*
- *Topography*
- *Hydraulic Properties,*
- *Subsurface Utilities*

Surface Water Flow System

- *Natural Drainage Profiles and Sections*
- *Drainage Diversions/Berms*
- *Hydraulic Structures - Dams, Weirs, Gates, others*

Subsurface

- *Groundwater Heads/Gradients*
- *Groundwater Flows*
- *Infiltration/Recharge*
- *Groundwater Discharge to Surface Water - Seepage*
- *Soil Moisture w/depth & time*

Surface Water

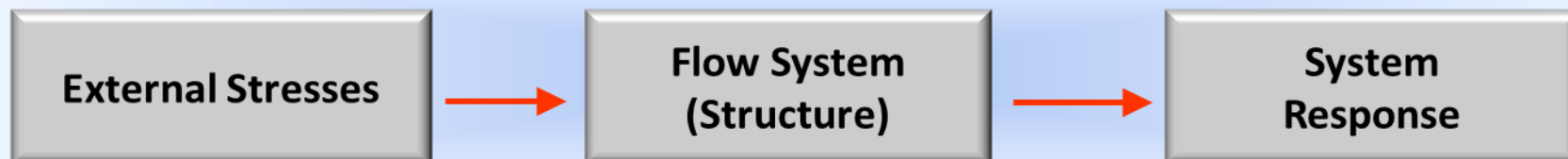
- *Overland Flow/Stage*
- *Surface Flow/Stage*
 - *Drainages*
 - *Structures*
- *Baseflows (Gaining/Losing)*
- *Pond/Reservoir levels*

Groundwater levels needed for calibration

- *Land Use (Pavement, bldgs)*

- *Infiltration Rates*
- *Evapotranspiration*

Flow Model



• Climate Data:

- Rainfall
- Reference ET
 - Wind
 - Air Temperature
 - Humidity
 - Solar Radiation

• Groundwater Pumping

- Irrigation
- Dewatering

• Artificial Recharge

• River Operations

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Surface Water

- Overland Flow/Stage
- Surface Flow/Stage
 - Drainages
 - Structures
- Baseflows (Gaining/Losing)
- Pond/Reservoir levels
- Infiltration Rates
- Evapotranspiration

Groundwater levels needed for calibration

Surface water flows needed for calibration

Data limitations

Never enough data!

- Interdisciplinary modelling requires interdisciplinary data
- Data is expensive – jurisdictional conflicts
- GIS and remote sensing revolutionizing data availability

Sources?

- National weather services
- Forestry and Agriculture agencies
- Surface water / flood protection agencies
- National geology/mining institutes
- Building and construction industry

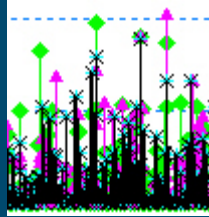
When do you have enough data?

04.

MIKE SHE User Interface

Data centric

Time
series
data



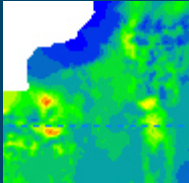
e.g. rainfall

Databases

End day	LAI	Root	Kc
90	2	1.2	0.55
180	4	1.3	0.55
270	5.5	1.7	0.55
365	3	1.2	0.55

e.g. vegetation info

Gridded
data



e.g. topography

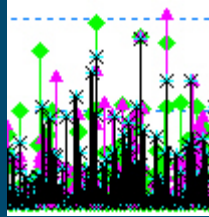
GIS/map
data



e.g. land use

Model linkages

Time
series
data



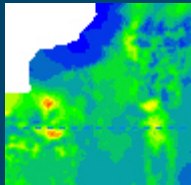
Databases

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90	2	1.2	0.55
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365	3	1.2	0.55



MIKE 11
(Surface water)

Gridded
data

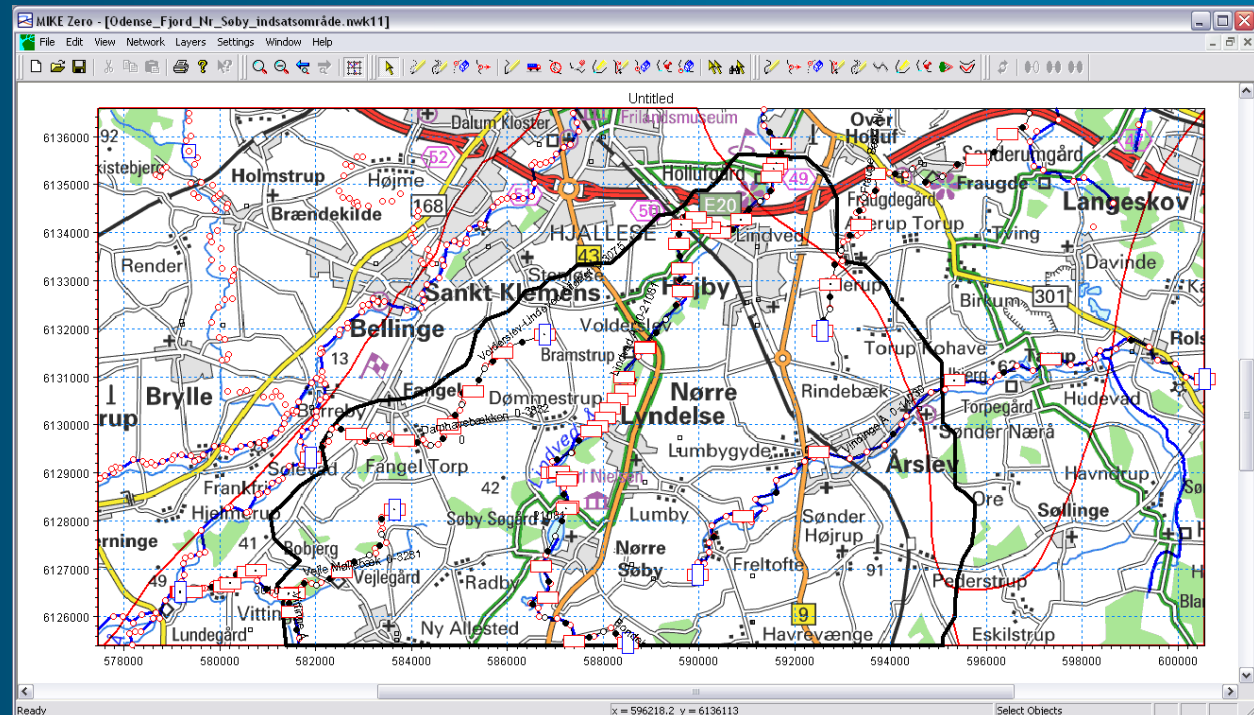


Mouse
(sewers)

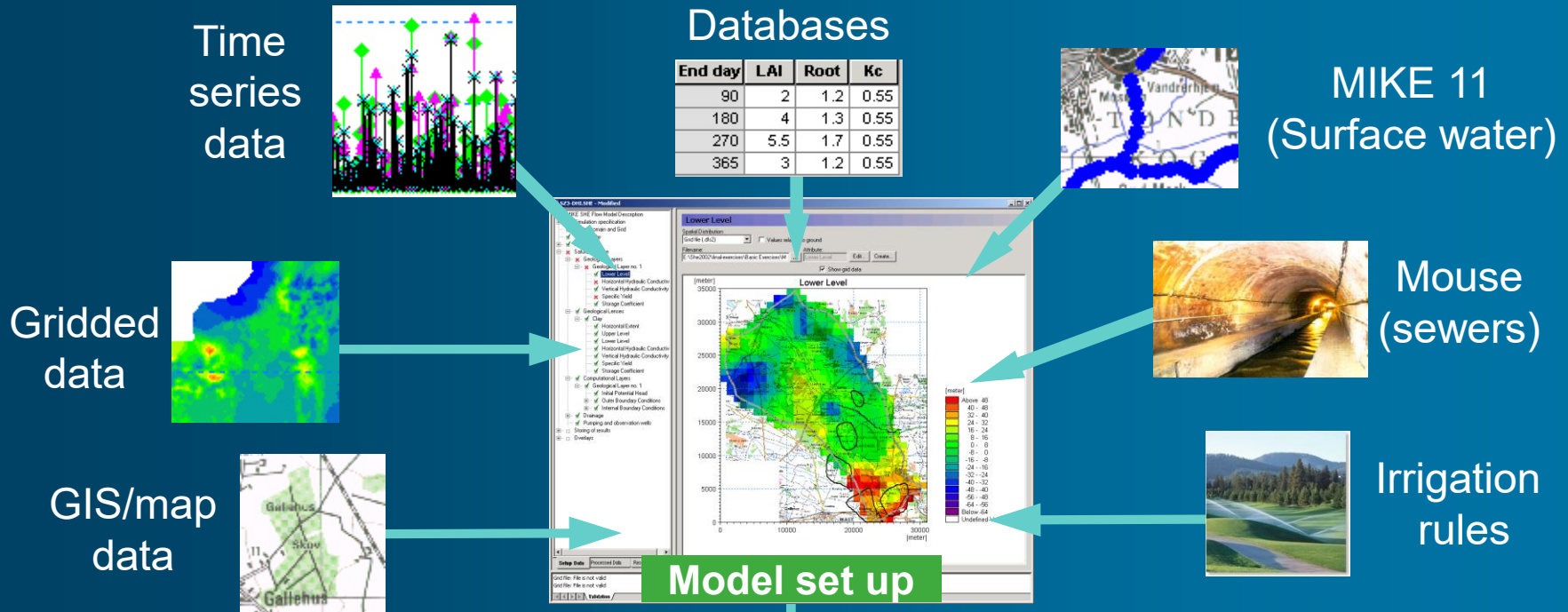
GIS/map
data



MIKE 11 Network editor



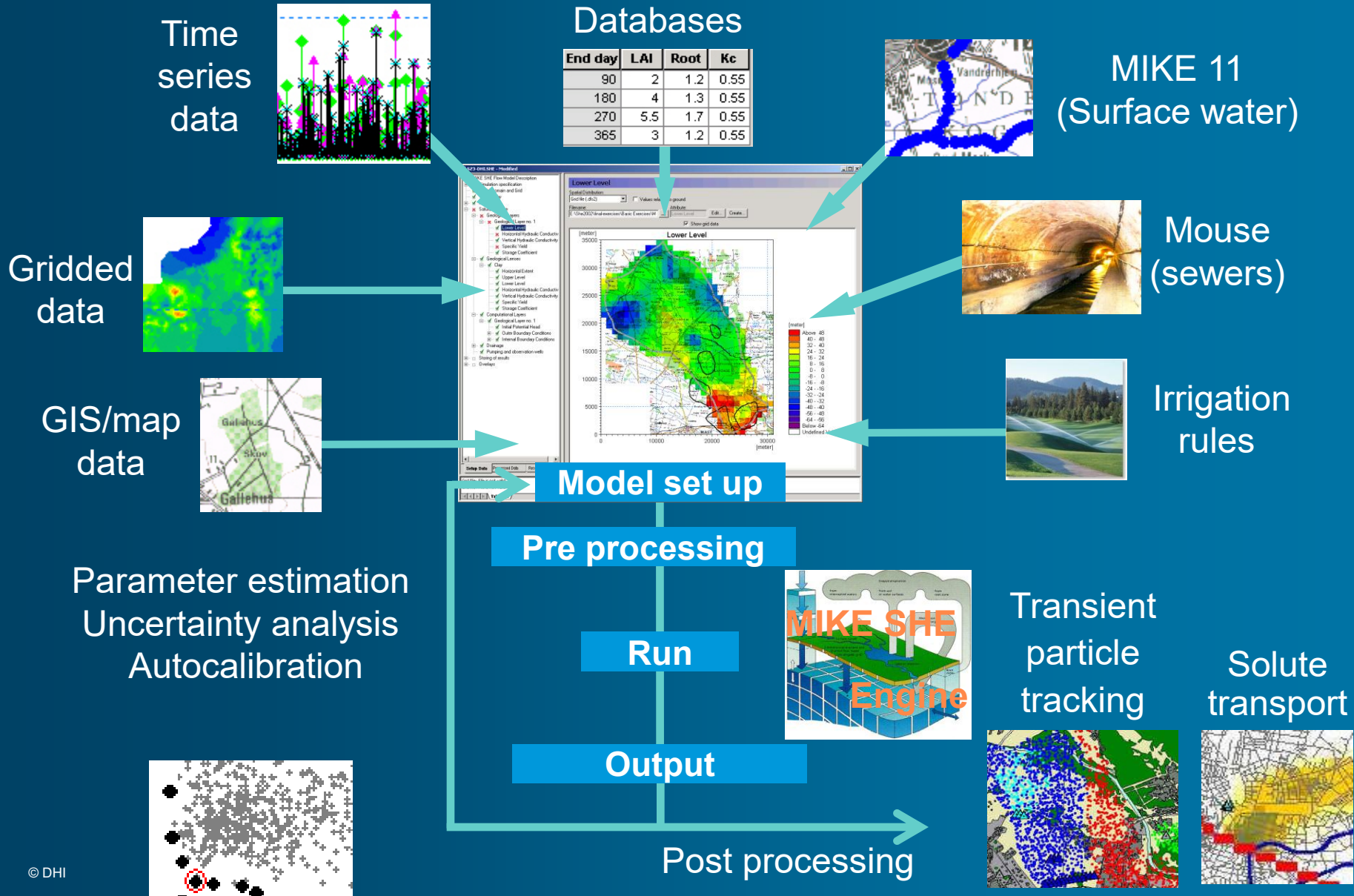
A modelling framework



- Model independent data entry
- Numerical model generated at run time
- Comprehensive output tools

Stream Hydrographs Time plots Map plots Calibration statistics Complete water balance

A modelling framework



User interface

Data centered structure preserves your data integrity and investment

Dynamic model browser and data tree

- Logical and intuitive work flow
- Flexible, fully scalable solution
- Model independent conceptual model

